

PHYS 202: Homework 11

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Problem 01

$$\begin{aligned}F_z &= \mu_z \alpha \\t &= \frac{L}{v} \\d &= \frac{1}{2} \left(\frac{F_z}{m} \right) t^2 = \frac{1}{2} \left(\frac{\mu_z \alpha}{m} \right) \left(\frac{L}{v} \right)^2 \implies \alpha = \frac{2mv^2 d}{\mu_z L^2} = \boxed{0.386}\end{aligned}$$

Problem 02

(a)

$$\begin{aligned}p &\sim \frac{\hbar}{r} = 1.05 \times 10^{-16} \text{ kg m/s} \\v &= p/m = 1.2 \times 10^{14} \text{ m/s} \gg 3.0 \times 10^8 \text{ m/s} = c\end{aligned}$$

This speed is extreme and un-relativistic thus unreal.

(b)

$$E \sim pc = \frac{\hbar c}{r} = 2 \times 10^5 \text{ MeV}$$

But rest energy of electron is 0.511 MeV. The classically flawed energy ratio is

$$\frac{E}{0.511} = 4 \times 10^5$$

This is unrealistically high proportion of energy.

Problem 03

(a)

Comparing with $n^2 \hbar^2 \pi^2 / 2mL^2$ we see that

$$n_1^2 + n_2^2 = 5$$

The only valid pairs that solve this above equation are

$$n_1 = 1, n_2 = 2$$

or

$$n_2 = 2, n_1 = 1$$

So the occupied states are $n = 1, 2$.

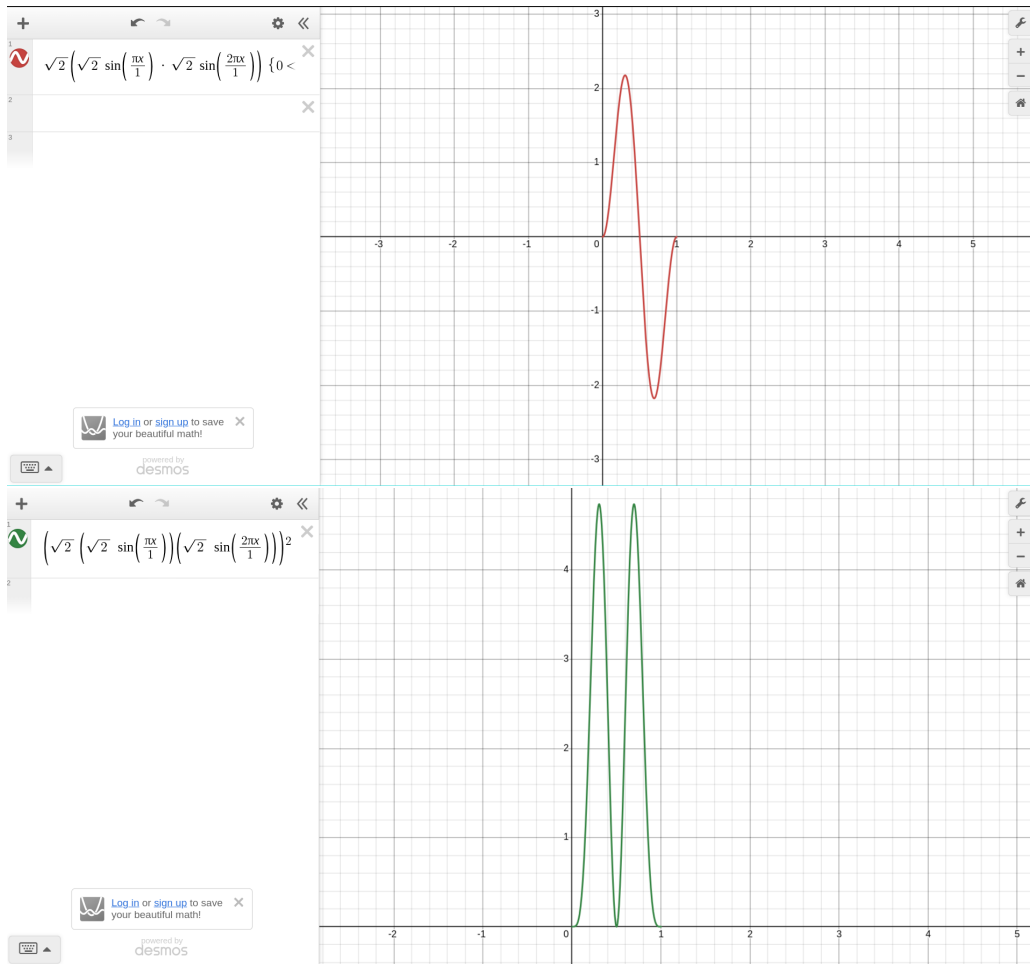
(b)

$$\psi_S(x_1, x_2) = \frac{1}{\sqrt{2}} (\psi_1(x_1)\psi_2(x_2) + \psi_2(x_1)\psi_1(x_2)) \implies \sqrt{2}\psi_1(x_1)\psi_2(x_1)$$

Wave functions are given for n state by

$$\sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$$

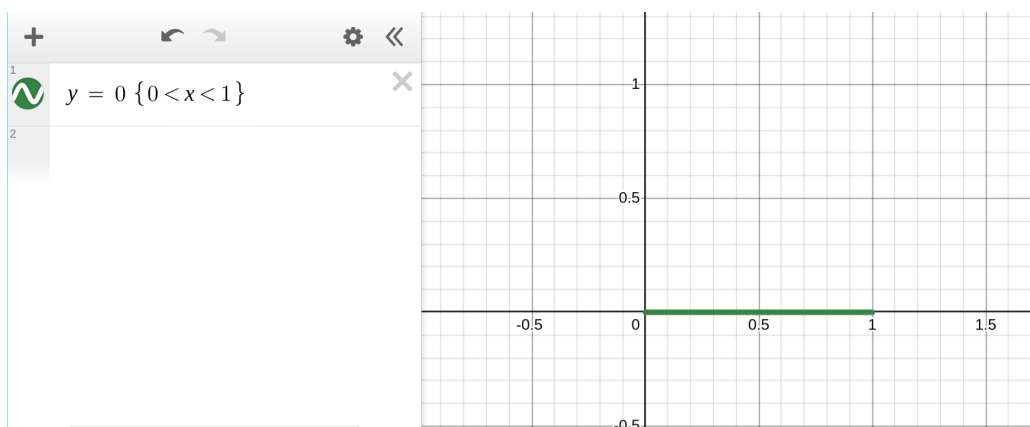
Plotting in desmos for $L = 1$ for $n = 1, 2$



Figur 1: Wave Function and Probability Plot

(c)

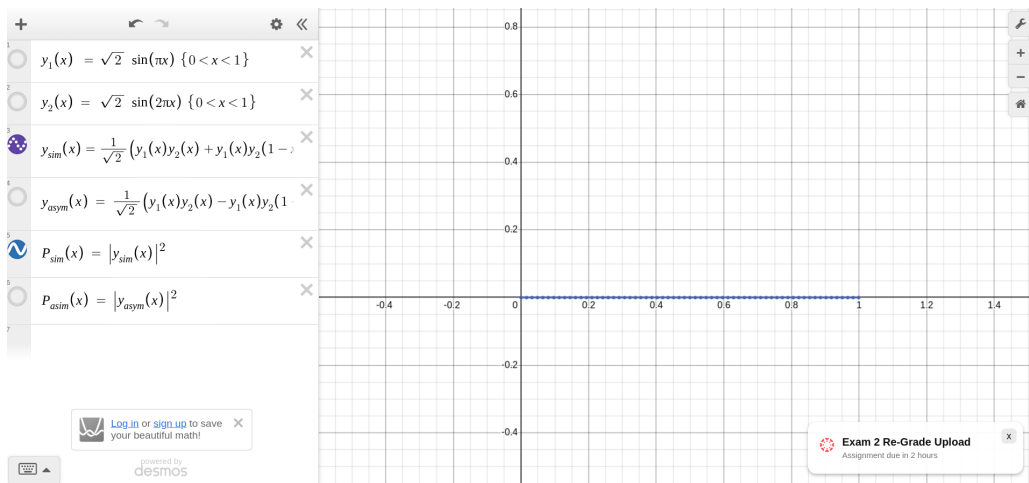
$$\psi_A(x_1, x_2) = \frac{1}{\sqrt{2}} (\psi_1(x_1)\psi_2(x_2) - \psi_2(x_1)\psi_1(x_2)) \underset{x_1=x_2}{\Rightarrow} 0$$



Figur 2: Wave Function (and so as probability)

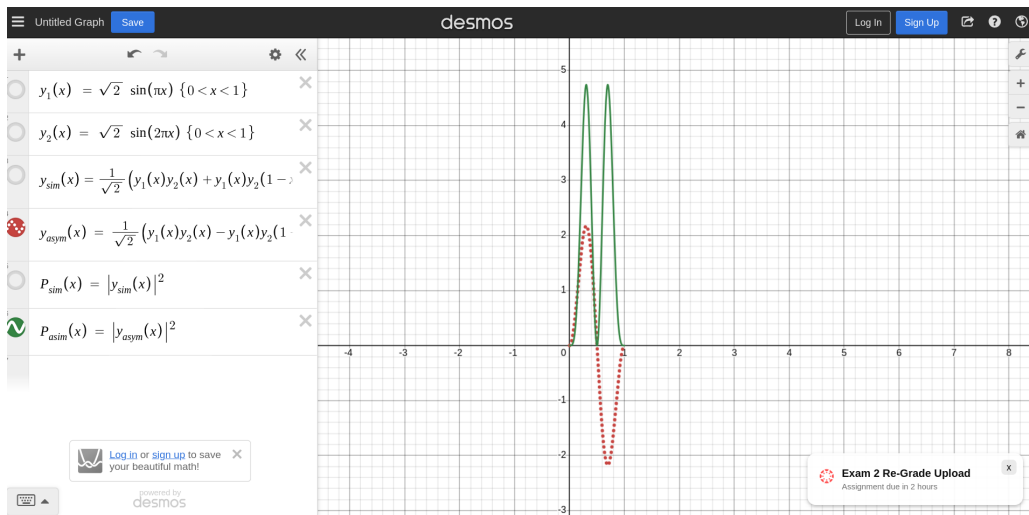
(d)

Symmetric wave function plot with $L - x$



Figuur 3: ./pics/4.png

Asymmetric Wave Function and Probability for $L - x$



Figuur 4: ./pics/5.png

Problem 04

Spin Half

Fermions are spin half. So each energy state can hold two of them for the two eigenvalues of the spins. For this

$$E_1 \implies 2$$

$$E_2 \implies 2$$

$$E_3 \implies 1$$

Total energy

$$E = \frac{\pi^2 \hbar^2}{2mL^2} (2(1^2) + 2(2^2) + 3^2) = \boxed{19 \frac{\pi^2 \hbar^2}{2mL^2}}$$

Spin one

Spin 1 are bosons so they all can take ground state for minimum energy

$$E = 5 \frac{\pi^2 \hbar^2}{2mL^2}$$

Spin 3/2

Again fermions, but this time the eigenvalues of spin are

$$\frac{3}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{3}{2}$$

So 4 of the lowest energy states can be filled up with 4 eigenvalues of the spin and the rest remaining is the $n = 2$ state hence we have $4 + (2)^2 = 8$

$$E = 8 \frac{\pi^2 \hbar^2}{2mL^2}$$

Problem 05

(a)

Swap for spatial part in state I

$$I = \psi_n(x_1) \uparrow \psi_{n'}(x_2) \downarrow - \psi_{n'}(x_1) \downarrow \psi_n(x_2) \uparrow \implies \psi_{n'}(x_1) \uparrow \psi_n(x_2) \downarrow - \psi_n(x_1) \downarrow \psi_{n'}(x_2) \uparrow = -II$$

Spatial state swap for I gives us $-II$. This also implies that swapping the states of II will give us $-I$.

Now we cannot call this either symmetric or antisymmetric because the swap is not positive or negative of it self - rather we have a totally different state (as we saw swapping I gives $-II$).

To be able to call I antisymmetric, we need the swap to give us $-I$ state, not $-II$.

Neither.

(b)

Swap the spin part of states,

$$I = \psi_n(x_1) \uparrow \psi_{n'}(x_2) \downarrow - \psi_{n'}(x_1) \downarrow \psi_n(x_2) \uparrow \implies \psi_n(x_1) \downarrow \psi_{n'}(x_2) \uparrow - \psi_{n'}(x_1) \uparrow \psi_n(x_2) \downarrow = II$$

Spin swaps for I gave us II states, hence it's obvious that swapping spin states for II would give us I state. Hence again, because the swapped states are neither $+$ or $-$ of themselves, we are left with the exchange symmetry being neither symmetric or antisymmetric.

Neither.

(c)

$$I + II = (\psi_n(x_1) \uparrow \psi_{n'}(x_2) \downarrow - \psi_{n'}(x_1) \downarrow \psi_n(x_2) \uparrow) + (\psi_n(x_1) \downarrow \psi_{n'}(x_2) \uparrow - \psi_{n'}(x_1) \uparrow \psi_n(x_2) \downarrow)$$

Notationally making a change,

$$\begin{aligned} & (\psi_n(x_1)\psi_{n'}(x_2) \uparrow\downarrow - \psi_{n'}(x_1)\psi_n(x_2) \downarrow\uparrow) + (\psi_n(x_1)\psi_{n'}(x_2) \downarrow\uparrow - \psi_{n'}(x_1)\psi_n(x_2) \uparrow\downarrow) \\ & (\psi_n(x_1)\psi_{n'}(x_2)) (\uparrow\downarrow + \downarrow\uparrow) - \psi_{n'}(x_1)\psi_n(x_2) (\downarrow\uparrow + \uparrow\downarrow) \\ & \boxed{[\psi_n(x_1)\psi_{n'}(x_2) - \psi_{n'}(x_1)\psi_n(x_2)] (\downarrow\uparrow + \uparrow\downarrow)} \end{aligned}$$

(d)

Spatial swap

$$\begin{aligned} & [\psi_n(x_1)\psi_{n'}(x_2) - \psi_{n'}(x_1)\psi_n(x_2)] (\downarrow\uparrow + \uparrow\downarrow) \implies [\psi_{n'}(x_1)\psi_n(x_2) - \psi_n(x_1)\psi_{n'}(x_2)] (\downarrow\uparrow + \uparrow\downarrow) \\ & \implies -[\psi_n(x_1)\psi_{n'}(x_2) - \psi_{n'}(x_1)\psi_n(x_2)] (\downarrow\uparrow + \uparrow\downarrow) \end{aligned}$$

Quite obviously

Spatially Antisymmetric

Spin Swap

$$\begin{aligned} & [\psi_n(x_1)\psi_{n'}(x_2) - \psi_{n'}(x_1)\psi_n(x_2)] (\downarrow\uparrow + \uparrow\downarrow) \implies [\psi_n(x_1)\psi_{n'}(x_2) - \psi_{n'}(x_1)\psi_n(x_2)] \underbrace{(\uparrow\downarrow + \downarrow\uparrow)}_{\text{look here bro}} \\ & \implies [\psi_n(x_1)\psi_{n'}(x_2) - \psi_{n'}(x_1)\psi_n(x_2)] (\downarrow\uparrow + \uparrow\downarrow) \end{aligned}$$

We got the same thing

Spininically Symmetric

(e)

Spin exchange symmetry implies unchanged states. Hence if we change both spatial and spin part, then only spatial acts to make the whole state negative hence keeping it anti-symmetric.

Still Anti-symmetric.

(f)

Repulsion requires the particles to be kept farther apart. Now in state *I* and *II* we have such case that $x_1 = x_2$ can still give non-zero probability of existing, whereas *sum* case give zero probability for $x_1 = x_2$. Hence the *sum* case more likely because repulsion makes it less likely for the particles to be at the same position (or close).

Problem 06

$$\begin{aligned}\psi_a &= \psi_{1,0,0}(\mathbf{r}) \uparrow \\ \psi_b &= \psi_{1,0,0}(\mathbf{r}) \downarrow \\ \psi_c &= \psi_{2,0,0}(\mathbf{r}) \uparrow\end{aligned}$$

Slater Determinant is written

$$\Psi(\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3) = \frac{1}{\sqrt{6}} \begin{vmatrix} \psi_{1,0,0}(\mathbf{r}_1) \uparrow & \psi_{1,0,0}(\mathbf{r}_2) \uparrow & \psi_{1,0,0}(\mathbf{r}_3) \uparrow \\ \psi_{1,0,0}(\mathbf{r}_1) \downarrow & \psi_{1,0,0}(\mathbf{r}_2) \downarrow & \psi_{1,0,0}(\mathbf{r}_3) \downarrow \\ \psi_{2,0,0}(\mathbf{r}_1) \uparrow & \psi_{2,0,0}(\mathbf{r}_2) \uparrow & \psi_{2,0,0}(\mathbf{r}_3) \uparrow \end{vmatrix}$$

The incredibly painful work of expanding this thing

$$\begin{aligned}\Psi = \frac{1}{\sqrt{6}} & \left(\psi_{1,0,0}(\mathbf{r}_1) \uparrow \cdot \psi_{1,0,0}(\mathbf{r}_2) \downarrow \cdot \psi_{2,0,0}(\mathbf{r}_3) \uparrow - \psi_{1,0,0}(\mathbf{r}_1) \uparrow \cdot \psi_{2,0,0}(\mathbf{r}_2) \uparrow \cdot \psi_{1,0,0}(\mathbf{r}_3) \downarrow + \psi_{1,0,0}(\mathbf{r}_2) \uparrow \cdot \psi_{2,0,0}(\mathbf{r}_3) \uparrow \cdot \psi_{1,0,0}(\mathbf{r}_1) \downarrow \right. \\ & \left. - \psi_{1,0,0}(\mathbf{r}_2) \downarrow \cdot \psi_{1,0,0}(\mathbf{r}_3) \uparrow \cdot \psi_{2,0,0}(\mathbf{r}_1) \uparrow + \psi_{1,0,0}(\mathbf{r}_3) \uparrow \cdot \psi_{1,0,0}(\mathbf{r}_1) \downarrow \cdot \psi_{2,0,0}(\mathbf{r}_2) \uparrow - \psi_{1,0,0}(\mathbf{r}_3) \uparrow \cdot \psi_{2,0,0}(\mathbf{r}_1) \uparrow \cdot \psi_{1,0,0}(\mathbf{r}_2) \downarrow \right)\end{aligned}$$

I counted 6 terms which is correct.

Problem 07

(a)

$$B = \frac{\mu_0 I}{2a} = \mu_0 \left(\frac{\mu_e}{\pi a^2} \right) \frac{1}{2a} = \frac{\mu_0}{2a} \frac{e\hbar}{2m_e \pi a^2} = \frac{\mu_0 e \hbar}{4\pi m_e a^3}$$

(b)

It's a very basic fact in condensed matter physics that anti-aligned configuration is lower in energy. In a bar magnet idea too, north and north repel so it's more sustainable to have an opposite configuration hence anti-aligned is lower in energy (as it is stable).

(c)

$$\Delta E = 2\mu_p B = 2 \left(5.6 \frac{e\hbar}{2m_p} \right) \left(\frac{\mu_0 e \hbar}{4\pi m_e a^3} \right) = 5.8 \times 10^{-6} \text{ eV}$$



Figuur 5: ./pics/6.png

(d)

$$E = \frac{hc}{\lambda} \implies \lambda = \frac{hc}{\Delta E}$$

Calculator gives

$$E = 0.21m = \boxed{21cm}$$

Problem 08

(a)

Eigenvalues are (PHYS 311/312)

$$\frac{3}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{3}{2}$$

(b)

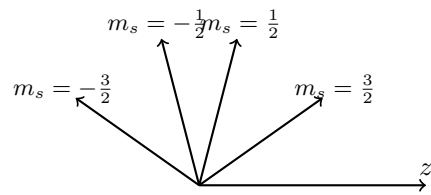
$$\cos \theta = \frac{S_z}{|\vec{S}|} = \frac{m_s \hbar}{\sqrt{s(s+1)} \hbar} = \frac{m_s}{\sqrt{s(s+1)}}$$

$$m_s = \frac{3}{2} \Rightarrow \cos \theta = \frac{3/2}{\sqrt{15}/2} = \frac{3}{\sqrt{15}} \Rightarrow \theta = \cos^{-1} \left(\frac{3}{\sqrt{15}} \right) \approx 35.26^\circ$$

$$m_s = \frac{1}{2} \Rightarrow \cos \theta = \frac{1}{\sqrt{15}} \Rightarrow \theta = \cos^{-1} \left(\frac{1}{\sqrt{15}} \right) \approx 75.52^\circ$$

$$m_s = -\frac{1}{2} \Rightarrow \cos \theta = -\frac{1}{\sqrt{15}} \Rightarrow \theta = \cos^{-1} \left(-\frac{1}{\sqrt{15}} \right) \approx 104.48^\circ$$

$$m_s = -\frac{3}{2} \Rightarrow \cos \theta = -\frac{3}{\sqrt{15}} \Rightarrow \theta = \cos^{-1} \left(-\frac{3}{\sqrt{15}} \right) \approx 144.74^\circ$$



Spin- $\frac{3}{2}$: Angles with the z -axis

(c)

Degeneracies are 4 fold for 4 possible eigenvalues of spin.